AFKEBOOKS
SINCE 2013

and Asia, transforming the recognition and management of incident tuberculosis and multidrug resistance where they are most prevalent. Since rifampin resistance frequently accompanies resistance to other antibiotics, this test can suggest the presence of multidrug-resistant *M. tuberculosis* within hours instead of weeks, without the infrastructure required for culture.

High-resolution genomic tracking of the spread of epidemics—from cholera to Ebola to Zika to COVID-19—has yielded insights into which public health measures may prove most effective in controlling local epidemics. Many genomic tracking efforts have involved close collaborations with local scientists and public health officials, and considerable investment in sequencing infrastructure in sub-Saharan Africa has made on-location epidemic tracking in the event of another such outbreak feasible. Such investment can not only enable real-time outbreak recognition and tracking but also provide the infrastructure needed to capitalize on the many other benefits of high-throughput sequencing as they are developed. The early returns of such investments are exemplified by the rapid reporting of genome sequences for SARS-CoV-2, with 55,000 viral genome sequences reported within the first 6 months of the pandemic. Overall, sequencing efforts have become cheaper and have moved closer to point-of-care with each passing year. As these technologies synergize with efforts to globalize information-technology resources, global implementation of genomic methods promises to spread state-of-the-art methods for diagnosis, treatment, and epidemic tracking of infections to areas that need these capabilities the most.

GENOMICS AND THE COVID-19 PANDEMIC

The COVID-19 pandemic, which began in 2019 and spread worldwide in 2020, resulted in hundreds of millions of documented infections and millions of deaths and serves as a prime example of the pandemic potential of infectious pathogens. It also demonstrated the central role that genomic tools now play in response to infectious outbreaks, ranging from enabling diagnostics and vaccines to tracking evolution, virulence, and transmissibility of the pathogen. The rapid discovery of SARS-CoV-2 and sequencing of its genome was complete within weeks of the recognition of the clinical syndrome. The rapid public sharing of this genome sequence led directly to two key interventions: diagnostic assay development via RT-qPCR and vaccine design. Crucially, vaccine development was informed by homology of the SARS-CoV-2 sequence to SARS and MERS coronaviruses. The dominant antigen of those viruses, the surface protein Spike, was well characterized, enabling the design of the first SARS-CoV-2 vaccines to begin the day after the genome sequence was shared. The progress of the most rapidly developed and validated vaccine in human history was unquestionably accelerated by genomic technology. WGS has also played a large role in outbreak tracking and confirmation of case clusters in institutional settings such as hospitals or congregate living facilities, in helping to distinguish reinfections from recrudescence or prolonged viral shedding, in monitoring spread through societies, and in tracking pathogen evolution, including the emergence of new variants of concern with altered transmissibility, severity, and/or partial evasion of the immune response generated to prior versions of the virus, vaccines, or monoclonal antibody therapeutics. Finally, cutting edge genomic methods including single-cell transcriptional profiling and genome-wide association studies are contributing to our understanding of the wide variability in outcomes of SARS-CoV-2 infection, ranging from asymptomatic carriage to death. Overall, just as the global response to the COVID-19 pandemic underscores the indispensable role that genomics methods have come to play in the clinical and public health management of infectious diseases, the devastating impact of this pandemic reveals the urgent need for further development and implementation of tools for disease surveillance and response.

SUMMARY

By illuminating the genetic information that encodes the most fundamental processes of life, genomic technologies are transforming many aspects of medicine. In infectious diseases, methods such as next-generation sequencing and genome-scale expression analysis offer information of unprecedented depth about individual microbes as well as microbial communities. This information is expanding our understanding of the interactions of microorganisms with each other, their human hosts, and the environment. Despite technological and financial barriers that continue to slow the widespread adoption of large-scale pathogen sequencing in clinical and public health settings, genomic methodologies have utterly transformed the research landscape in infectious disease and are beginning to make meaningful inroads into clinical settings. As even vaster amounts of data are generated, innovations in data storage, development of bioinformatics tools to manipulate the data, standardization of methods, and training of end-users in both the research and clinical realms will be required. The cost-effectiveness and applicability of whole genome sequencing, particularly in the clinic, remain to be studied, and studies of the impact of genome sequencing on patient outcomes will be needed to clarify the contexts in which these new methodologies can make the greatest contributions to patient well-being. The ongoing efforts to overcome limitations through collaboration, teaching, and reduction of financial obstacles should be applauded and expanded. With advances in genomic technologies and computational analysis, our ability to detect, characterize, treat, monitor, prevent, and control infections has advanced rapidly in recent years and will continue to do so, with the hope of heralding a new era where the clinician is better armed to combat infection and promote human health.

FURTHER READING

Bullman S et al: Emerging concepts and technologies for the discovery of microorganisms involved in human disease. Annu Rev Pathol 12:217, 2017.

Burnham CD et al: Diagnosing antimicrobial resistance. Nat Rev Microbiol 15:697, 2017.

Croucher NJ et al: Population genomics of post-vaccine changes in pneumococcal epidemiology. Nat Genet 45:656, 2013.

Cryptic Consortium et al: Prediction of susceptibility to first-line tuberculosis drugs by DNA sequencing. N Engl J Med 379:1403, 2018.

Dudas G et al: Virus genomes reveal factors that spread and sustained the Ebola epidemic. Nature 544:309, 2017.

Gardy JL, Loman NJ: Towards a genomics-informed, real-time, global pathogen surveillance system. Nat Rev Genet 19:9, 2018.

Grubaugh ND et al: Tracking virus outbreaks in the twenty-first century. Nat Microbiol 4:10, 2019.

Loman NJ, Pallen MJ: Twenty years of bacterial genome sequencing. Nat Rev Microbiol 13:787, 2015.

Mutreja A et al: Evidence for several waves of global transmission in the seventh cholera pandemic. Nature 477:462, 2011.

Wu Z, McGoogan JM: Characteristics of and important lessons from the coronavirus disease 2019 (COVID-19) outbreak in China. JAMA 323:1239, 2020.

122 Approach to the Acutely Ill Infected Febrile Patient

Tamar F. Barlam

The physician treating the acutely ill febrile patient must be able to recognize infections that require emergent attention. If such infections are not adequately evaluated and treated at initial presentation, the opportunity to alter an adverse outcome may be lost. In this chapter, the clinical presentations of and approach to patients with infectious disease emergencies are discussed. These infectious processes and their treatments are discussed in detail in other chapters.

AFKEBOOKS
SINCE 2013

APPROACH TO THE PATIENT

Acute Febrile Illness

Before the history is elicited and a physical examination is performed, an immediate assessment of the patient's general appearance can yield valuable information. The perceptive physician's subjective sense that a patient is septic or toxic often proves accurate. Visible agitation or anxiety in a febrile patient can be a harbinger of critical illness.

HISTORY

Presenting symptoms are frequently nonspecific. Detailed questions should be asked about the onset and duration of symptoms and about changes in severity or rate of progression over time. Host factors, such as extremes of age, and comorbid conditions may increase the risk of infection with certain organisms or of a more fulminant course than is usually seen. Lack of splenic function, alcoholism with significant liver disease, IV drug use, HIV infection, diabetes, malignancy, morbid obesity, organ transplantation, and chemotherapy all predispose to specific infections and frequently to increased severity. The patient should be questioned about factors that might help identify a nidus for invasive infection, such as recent upper respiratory tract infections, influenza, or varicella; prior trauma; disruption of cutaneous barriers due to lacerations, burns, surgery, body piercing, or decubiti; and the presence of foreign bodies or prosthetic devices. Travel, presence during a natural disaster such as a hurricane or tsunami, contact with pets or other animals, or activities that might result in tick or mosquito exposure can lead to diagnoses that would not otherwise be considered. Recent dietary intake, medication use, social or occupational contact with ill individuals, vaccination history, recent sexual contacts, and menstrual history may be relevant. Pregnancy might increase the risk and severity of some illnesses, such as influenza or COVID-19, or increase the risk of significant morbidity for the fetus, as in *Listeria* or Zika virus infection. A detailed review of systems should include any neurologic signs or sensorium alterations, rashes or skin lesions, and focal pain or tenderness.

PHYSICAL EXAMINATION

A complete physical examination should be performed, with special attention to several areas that are sometimes given short shrift in routine examinations such as assessment of the patient's general appearance and a detailed skin, soft tissue and neurologic evaluation.

The patient may appear either anxious and agitated or lethargic and apathetic. Fever is usually present, although elderly patients and compromised hosts (e.g., patients who are uremic or cirrhotic and those who are taking glucocorticoids or nonsteroidal anti-inflammatory drugs) may be afebrile despite serious underlying infection. Critically ill patients may be hypothermic, with a high risk of organ failure and mortality. Mortality at 30 days has been shown to decrease with increasing temperature at presentation. Measurement of blood pressure, heart rate, and respiratory rate and oxygen saturation helps determine the degree of hemodynamic and metabolic compromise. The patient's airway must be evaluated to rule out the risk of obstruction from an invasive oropharyngeal infection.

The etiologic diagnosis may become evident in the context of a thorough skin examination **(Chap. 19)**. Petechial rashes are typically seen with meningococcemia or Rocky Mountain spotted fever (RMSF; **see Fig. A1-16**); erythroderma is associated with toxic shock syndrome (TSS). On soft tissue and muscle examination, areas of erythema or duskiness, edema, and tenderness may indicate underlying necrotizing fasciitis, myositis, or myonecrosis. The neurologic examination must include a careful assessment of mental status for signs of early encephalopathy. Evidence of nuchal rigidity or focal neurologic findings should be sought.

DIAGNOSTIC WORKUP

After a quick clinical assessment, diagnostic material should be obtained rapidly and antibiotic and supportive treatment begun. Blood (for cultures; baseline complete blood count with differential; measurement of serum electrolytes, blood urea nitrogen, serum creatinine, and serum glucose; liver function tests and serum lactate; C-reactive protein, lactate dehydrogenase, and D-dimer) can be obtained at the time an IV line is placed and before antibiotics are administered. Three sets of blood cultures should be performed for patients with possible acute endocarditis. Blood smears from patients at risk for severe parasitic disease, such as malaria or babesiosis **(Chaps. 224, 225, and A2)**, must be examined for the diagnosis and quantitation of parasitemia. Blood smears may also be diagnostic in ehrlichiosis and anaplasmosis. Testing of a nasopharyngeal sample for possible COVID-19 may be indicated.

Patients with possible meningitis should have cerebrospinal fluid (CSF) drawn before the initiation of antibiotic therapy. Focal findings, depressed mental status, or papilledema should be evaluated by brain imaging prior to lumbar puncture, which, in this setting, could initiate herniation. *Antibiotics should be administered before imaging but after blood for cultures has been drawn.* If CSF cultures are negative, blood cultures will provide the diagnosis in 50–70% of cases. Molecular diagnostic techniques (e.g., broad-range 16S rRNA gene polymerase chain reaction testing for bacterial meningitis pathogens) are of increasing importance in the rapid diagnosis of life-threatening infections.

Focal abscesses necessitate immediate CT or MRI as part of an evaluation for surgical intervention. Other diagnostic procedures, such as wound cultures, should not delay the initiation of treatment for more than minutes. Once emergent evaluation, diagnostic procedures, and (if appropriate) surgical consultation (see below) have been completed, other laboratory tests can be conducted. Appropriate radiography, CT and/or MRI imaging, urinalysis, measurement of the erythrocyte sedimentation rate, C-reactive protein, and/or procalcitonin, and transthoracic or transesophageal echocardiography all may prove important.

TREATMENT

The Acutely Ill Patient

In the acutely ill patient, empirical antibiotic therapy for presumed bacterial or fungal infection is critical and should be administered without undue delay in addition to fluid resuscitation and vasopressor support as needed. Increased prevalence of antibiotic resistance in community-acquired bacteria must be considered when antibiotics are selected. **Table 122-1** lists first-line empirical regimens for infections considered in this chapter. In addition to the rapid initiation of antibiotic therapy, several of these infections require urgent surgical attention. Neurosurgical evaluation for subdural empyema, otolaryngologic surgery for possible mucormycosis, and cardiothoracic surgery for critically ill patients with acute endocarditis are as important as antibiotic therapy. For infections such as necrotizing fasciitis and clostridial myonecrosis, rapid surgical intervention supersedes other diagnostic or therapeutic maneuvers.

Adjunctive treatments may reduce morbidity and mortality rates and include dexamethasone for bacterial meningitis or IV immunoglobulin for TSS. Adjunctive therapies should usually be initiated within the first hours of treatment; however, dexamethasone for bacterial meningitis must be given before or at the time of the first dose of antibiotic. Glucocorticoids may also be harmful—e.g., when given in the setting of cerebral malaria or viral hepatitis.

SPECIFIC PRESENTATIONS

The infections considered below according to common clinical presentation can have rapidly catastrophic outcomes, and their immediate recognition and treatment can be life-saving. Recommended empirical therapeutic regimens are presented in Table 122-1.

SEPSIS WITHOUT AN OBVIOUS FOCUS OF PRIMARY INFECTION

Patients initially have a brief prodrome of nonspecific symptoms and signs that progresses quickly to hemodynamic instability with

AFKEBOOKS
SINCE 2013

TABLE 122-1 Empirical Treatment for Common Infectious Disease Emergencies[a]

CLINICAL SYNDROME	POSSIBLE ETIOLOGIES	TREATMENT	COMMENTS	SEE CHAP(S).
Sepsis without a Clear Focus				
Septic shock	*Pseudomonas* spp., gram-negative enteric bacilli, *Staphylococcus* spp., *Streptococcus* spp.	Vancomycin (15 mg/kg q12h)[b] ***plus either*** Piperacillin/tazobactam (4.5 g) q8h via extended infusion (EI)[c] *or* cefepime (2 g) q8h via EI	Empirical therapy should be tailored to local resistance patterns. Carbapenem or aminoglycoside antibiotics should be considered for empirical therapy when rates of multidrug-resistant gram-negative organisms are high or for patients with risk factors for resistant organisms. Adjust treatment when culture data become available.	147, 148, 161, 164, 304
Overwhelming post-splenectomy sepsis	*Streptococcus pneumoniae, Haemophilus influenzae, Neisseria meningitidis*	Ceftriaxone (2 g q12h) *plus* vancomycin (15 mg/kg q12h)[b]	If a β-lactam–sensitive strain is identified, vancomycin can be discontinued and a narrower-spectrum agent, such as penicillin, considered based on susceptibility testing.	304
Babesiosis	*Babesia microti* (U.S.), *B. divergens* (Europe)	Atovaquone (750 mg q12h) *plus* azithromycin (500 mg q24h)	Clindamycin (600 mg q8h) *plus* quinine (650 mg q8h) can be used in severe disease not responding to atovaquone and azithromycin. Treatment with doxycycline (100 mg bid) for potential co-infection with *Borrelia burgdorferi* or *Anaplasma* spp. may be prudent.	222, 225
Sepsis with Skin Findings				
Meningococcemia	*N. meningitidis*	Ceftriaxone (2 g q12h) or penicillin (4 mU q4h)	Ceftriaxone eradicates nasopharyngeal carriage of the organism. Close contacts require chemoprophylaxis with rifampin (600 mg q12h for 2 days) or ciprofloxacin (a single dose, 500 mg).	155
Rocky Mountain spotted fever (RMSF)	*Rickettsia rickettsii*	Doxycycline (100 mg bid)	If both meningococcemia and RMSF are being considered, use ceftriaxone (2 g q12h) *plus* doxycycline (100 mg bid). If RMSF is diagnosed, doxycycline is the proven superior agent.	187
Purpura fulminans	*S. pneumoniae, H. influenzae, N. meningitidis*	Ceftriaxone (2 g q12h) *plus* vancomycin (15 mg/kg q12h)[b]	If a β-lactam-sensitive strain is identified, vancomycin can be discontinued.	146, 155, 157, 304
Erythroderma: toxic shock syndrome	Group A *Streptococcus, Staphylococcus aureus*	Vancomycin (15 mg/kg q12h)[b] *plus* clindamycin (600 mg q8h)	If a penicillin- or oxacillin-sensitive strain is isolated, these agents are superior to vancomycin (penicillin, 2 mU q4h; or oxacillin, 2 g IV q4h). The site of toxigenic bacteria should be debrided; IV immunoglobulin can be used in severe cases.[d]	147, 148
Sepsis with Soft Tissue Findings				
Necrotizing fasciitis	Group A *Streptococcus,* mixed aerobic/anaerobic flora, CA-MRSA[e]	Vancomycin (15 mg/kg q12h)[b] *plus* piperacillin/tazobactam (4.5 q q8h via EI)[c] *plus* clindamycin (600 mg q8h)	Urgent surgical evaluation is critical. Empirical therapy should be tailored to local resistance patterns. For mixed aerobic/anaerobic infections, clindamycin can be discontinued. Adjust treatment when culture data become available.	129, 147, 148
Clostridial myonecrosis	*Clostridium perfringens*	Penicillin (2 mU q4h) *plus* clindamycin (600 mg q8h)	Urgent surgical evaluation is critical.	154
Neurologic Infections				
Bacterial meningitis	*S. pneumoniae, N. meningitidis*	Ceftriaxone (2 g q12h) *plus* vancomycin (15 mg/kg q12h)[b]	If a β-lactam–sensitive strain is identified, vancomycin can be discontinued. If the patient is >50 years old or has comorbid disease, add ampicillin (2 g q4h) for *Listeria* coverage. Dexamethasone (10 mg q6h for 4 days) started before, or at the time of, the first dose of antibiotic improves outcome in adults with meningitis (especially pneumococcal).	138
Brain abscess, suppurative intracranial infections	*Streptococcus* spp., *Staphylococcus* spp., anaerobes, gram-negative bacilli	Vancomycin (15 mg/kg q12h)[b] *plus* metronidazole (500 mg q8h) *plus* ceftriaxone (2 g q12h)	Urgent surgical evaluation is critical. If a penicillin- or oxacillin-sensitive strain is isolated, these agents are superior to vancomycin (penicillin, 4 mU q4h; *or* oxacillin, 2 g q4h).	138
Cerebral malaria	*Plasmodium falciparum*	Artesunate (2.4 mg/kg IV at 0, 12, and 24 h; then once daily)[f]	Avoid glucocorticoids. Until IV artesunate is available, treatment can be initiated with oral artemether-lumefantrine. Atovaquone-proguanil, quinine, and mefloquine are other options.	222, 224
Spinal epidural abscess	*Staphylococcus* spp., gram-negative bacilli	Vancomycin (15 mg/kg q12h)[b] ***plus either*** Piperacillin/tazobactam (4.5 g q8h via EI) *or* cefepime (2 g q8h via EI)[c]	Surgical evaluation is essential. If a penicillin- or oxacillin-sensitive strain is isolated, these agents are superior to vancomycin (penicillin, 4 mU q4h; *or* oxacillin, 2 g q4h).	442
Focal Infections				
Acute bacterial endocarditis	*S. aureus,* β-hemolytic streptococci, HACEK group,[g] *Neisseria* spp., *S. pneumoniae*	Ceftriaxone (2 g q12h) or cefepime (2 g q8h via EI)[c] *plus* vancomycin (15 mg/kg q12h)[b]	Adjust treatment when culture data become available. Surgical evaluation is essential.	128

[a]These empirical regimens include coverage for gram-positive pathogens that are resistant to β-lactam antibiotics. Local resistance patterns should be considered and may alter the need for empirical vancomycin or for expanded coverage for antibiotic-resistant gram-negative pathogens. [b]A vancomycin loading dose of 20–25 mg/kg can be considered in critically ill patients. Dosing must be adjusted based on pharmacokinetic/pharmacodynamic monitoring. [c]EI=extended infusion. β-Lactam antibiotics may exhibit unpredictable pharmacodynamics in sepsis. Prolonged or continuous infusions are often used. [d]The optimal dose of IV immunoglobulin has not been determined, but the median dose in observational studies is 2 g/kg (total dose administered for 1–5 days). [e]Community-acquired methicillin-resistant *S. aureus*. [f]In the United States, artesunate must be obtained through the Centers for Disease Control and Prevention. [g]*Haemophilus* spp., *Aggregatibacter* spp., *Cardiobacterium hominis, Eikenella corrodens*, and *Kingella kingae*.

AFKEBOOKS
SINCE 2013

hypotension, tachycardia, tachypnea, respiratory distress, and altered mental status. Disseminated intravascular coagulation (DIC) with clinical evidence of a hemorrhagic diathesis is a poor prognostic sign.

Septic Shock (See also Chap. 304) Patients with bacteremia leading to septic shock may have a primary site of infection (e.g., pneumonia, pyelonephritis, or cholangitis) that is not evident initially. Elderly patients who may have atypical presentations and often have comorbid conditions, hosts compromised by malignancy and neutropenia, and patients who have recently undergone a surgical procedure or hospitalization are at increased risk for an adverse outcome. Gram-negative bacteremia with organisms such as *Pseudomonas aeruginosa* or *Escherichia coli* and gram-positive infection with organisms such as *Staphylococcus aureus* (including methicillin-resistant *S. aureus* [MRSA]) or group A streptococci can present as intractable hypotension and multiorgan failure. Treatment can usually be initiated empirically on the basis of the presentation, host factors (Chap. 304), and local patterns of bacterial resistance. Outcomes are worse when antimicrobial treatment is delayed or when the responsible pathogen ultimately proves not to be susceptible to the initial regimen. The increasing prevalence of multidrug-resistant organisms makes this especially relevant. Broad-spectrum antimicrobial agents are therefore recommended and should be instituted rapidly, preferably within the first hours after presentation. Pharmacodynamics are altered in sepsis due to increased volume of distribution and renal clearance, so it is important to adequately dose antimicrobials. Risk factors for fungal infection should be assessed, as the incidence of fungal septic shock is increasing. Nonbacterial causes of shock, such as dengue virus infection, should be considered in endemic areas as mortality can increase if undiagnosed. Biomarkers such as C-reactive protein and procalcitonin have not proved reliable diagnostically or prognostically but, when measured over time, can facilitate appropriate de-escalation of therapy with improved outcomes. Glucocorticoids are often considered for patients with severe sepsis who do not respond to fluid resuscitation and vasopressor therapy, but conclusive evidence for efficacy in this setting is lacking.

Overwhelming Infection in Asplenic Patients (See also Chap. 304) Patients without splenic function are at risk for overwhelming bacterial sepsis. Asplenic adult patients succumb to sepsis at 58 times the rate of the general population. Most infections occur within the first l or 2 years, but the increased risk persists throughout life. The median interval between splenectomy and sepsis is 4–6 years, with a range of 1–19 years. In asplenia, encapsulated bacteria cause the majority of infections. Adults, who are more likely to have antibody to these organisms, are at lower risk than children. *Streptococcus pneumoniae* is the most common isolate, causing 40–70% of cases. The risk of infection with *Haemophilus influenzae* or *Neisseria meningitidis* also is greater in patients without splenic function, but reported cases are declining. Severe clinical manifestations of infections due to other organisms, such as *E. coli*, *S. aureus*, *Bordetella holmesii*, *Capnocytophaga*, *Babesia*, and *Plasmodium* species, have been described.

Babesiosis (See also Chap. 225) A history of recent travel to endemic areas raises the possibility of infection with *Babesia*. Between 1 and 4 weeks after a tick bite, the patient experiences chills, fatigue, anorexia, myalgia, arthralgia, shortness of breath, nausea, and headache; ecchymosis and/or petechiae are occasionally seen. The tick that most commonly transmits *Babesia*, *Ixodes scapularis*, also transmits *Borrelia burgdorferi* (the agent of Lyme disease) and *Anaplasma*; coinfection can occur, and may result in more severe disease. Infection with the European species *Babesia divergens* is more frequently fulminant than that due to the U.S. species *Babesia microti*. *B. divergens* causes a febrile syndrome with hemolysis, jaundice, hemoglobinemia, and renal failure and is associated with a mortality rate of >40%. Severe babesiosis is especially common in asplenic hosts but does occur in hosts with normal splenic function, particularly those >60 years of age and those with underlying immunosuppressive conditions such as HIV infection or malignancy. Complications include renal failure, acute respiratory failure, DIC, and splenic rupture.

Other Sepsis Syndromes Tularemia (Chap. 170) has been reported in every U.S. state except Hawaii. This disease is associated with wild rabbit, tick, horse-fly, and tabanid fly contact. It can be transmitted by arthropod bite, handling of infected animal carcasses, consumption of contaminated food and water, or inhalation. The typhoidal form can be associated with gram-negative septic shock and a mortality rate of >30%, especially in patients with underlying comorbid or immunosuppressive conditions. Plague occurs infrequently in the United States (Chap. 171), primarily after contact with ground squirrels, prairie dogs, or chipmunks, but is endemic in other parts of the world; >90% of all cases occur in Africa with Madagascar especially affected. The septic form is particularly rare and is associated with shock, multiorgan failure, and a 30% mortality rate. Pneumonic plague is rapidly progressive and fatal without treatment. These infections should be considered in the appropriate epidemiologic setting. The Centers for Disease Control and Prevention (CDC) lists *Francisella tularensis* and *Yersinia pestis* (the agents of tularemia and plague, respectively) along with *Bacillus anthracis* (the agent of anthrax) as important organisms that might be used for bioterrorism (Chap. S3).

SEPSIS WITH SKIN MANIFESTATIONS (SEE ALSO CHAP. 19)

Maculopapular rashes may reflect early meningococcal or rickettsial disease but are usually associated with nonemergent infections. Exanthems are usually viral. Primary HIV infection commonly presents with a rash that is typically maculopapular and involves the upper part of the body but can spread to the palms and soles. The patient is usually febrile and can have lymphadenopathy, severe headache, dysphagia, diarrhea, myalgias, and arthralgias. Recognition of this syndrome provides an opportunity to prevent transmission and to institute early treatment.

Petechial rashes caused by viruses are seldom associated with hypotension or a toxic appearance, although there can be exceptions (e.g., severe measles or arboviral infection). Petechial rashes limited to the distribution of the superior vena cava are rarely associated with severe disease. In other settings, petechial rashes require more urgent attention.

Meningococcemia (See also Chap. 155) Almost three-quarters of patients with *N. meningitidis* bacteremia have a rash. Meningococcemia most often affects young children (i.e., those 6 months to 5 years old). In sub-Saharan Africa, the high prevalence of serogroup A meningococcal disease has been a threat to public health for more than a century. Thousands of deaths occur annually in this area, which is known as the "meningitis belt," and large epidemic waves occur approximately every 8–12 years. Serogroups W135, X, and C also are important emerging pathogens in Africa. For example, Nigeria experienced a large serogroup C outbreak in 2016–2017 in the setting of aggressive vaccination programs against serogroup A. Outside Africa, outbreaks for the past 50 years reported in the United States and Europe are caused primarily by serogroup C (approximately 60%) followed by serogroup B (29%). In the United States, sporadic cases and outbreaks occur in day-care centers, schools (grade school through college, particularly among college freshmen living in residential halls), and army barracks. Household contacts of index cases are at 400–800 times greater risk of disease than the general population. Patients may have fever, headache, nausea, vomiting, myalgias, changes in mental status, and meningismus. However, the rapidly progressive form of disease is not usually associated with meningitis. The rash is initially pink, blanching, and maculopapular, appearing on the trunk and extremities, but then becomes hemorrhagic, forming petechiae. Petechiae are first seen at the ankles, wrists, axillae, mucosal surfaces, and palpebral and bulbar conjunctiva, with subsequent spread on the lower extremities and to the trunk. A cluster of petechiae may be seen at pressure points—e.g., where a blood pressure cuff has been inflated. In rapidly progressive meningococcemia (10–20% of cases), the petechial rash quickly becomes purpuric (see Fig. A1-41), and patients develop DIC, multiorgan failure, and shock; 50–60% of these patients die, and survivors often require extensive debridement or

AFKEBOOKS
SINCE 2013

amputation of gangrenous extremities. Hypotension with petechiae for <12 h is associated with significant mortality. Cyanosis, coma, oliguria, metabolic acidosis, and elevated partial thromboplastin time also are associated with a fatal outcome. Antibiotics given in the office by the primary care provider before hospital evaluation and admission may improve prognosis; this observation suggests that early initiation of treatment may be life-saving. Meningococcal conjugate vaccines are protective against serogroups A, C, Y and W135 and are recommended for children 11–12 years of age with a booster dose at 16 years of age, and for other high-risk patients. Vaccines active against serogroup B are recommended for high-risk individuals ≥10 years of age and may be appropriate for teens and young adults (16 through 23 years of age).

Rocky Mountain Spotted Fever and Other Rickettsial Diseases (See also Chap. 187) RMSF is a tickborne disease that occurs throughout North and South America. It is caused primarily by *Rickettsia rickettsii* but can be caused by other rickettsiae (e.g., *R. parkeri*, *R. akari*). Up to 40% of patients do not report a history of a tick bite, but a history of travel or outdoor activity (e.g., camping in tick-infested areas) can often be ascertained. For the first 3 days, headache, fever, malaise, myalgias, nausea, vomiting, and anorexia are documented. By day 3, half of patients have skin findings. Blanching macules develop initially on the wrists and ankles and then spread over the legs and trunk. The lesions become hemorrhagic and are frequently petechial. The rash spreads to palms and soles later in the course. The centripetal spread is a classic feature of RMSF but occurs in a minority of patients. Moreover, 10–15% of patients with RMSF never develop a rash. The patient can be hypotensive and develop noncardiogenic pulmonary edema, confusion, lethargy, and encephalitis progressing to coma. The CSF contains 10–100 cells/μL, usually with a predominance of mononuclear cells. The CSF glucose level is often normal; the protein concentration may be slightly elevated. Renal and hepatic injury as well as bleeding secondary to vascular damage are noted. Delayed recognition and treatment are associated with a greater risk of death; mortality rates are 20–30% if untreated. Native Americans, Alaskan natives, Pacific Islanders, children 5–9 years of age, adults >70 years old, and persons with underlying immunosuppression are at increased risk of death as well.

Other rickettsial diseases cause significant morbidity and mortality worldwide. *Mediterranean spotted fever* caused by *Rickettsia conorii* is found in Africa, southwestern and south-central Asia, and southern Europe. Patients have fever, flu-like symptoms, and an inoculation eschar at the site of the tick bite. A maculopapular rash develops within 1–7 days, involving the palms and soles but sparing the face. Elderly patients or those with diabetes, alcoholism, uremia, or congestive heart failure are at risk for severe disease characterized by neurologic involvement, respiratory distress, and gangrene of the digits or purpura fulminans. Mortality rates associated with this severe form of disease approach 50%. *Epidemic typhus*, caused by *Rickettsia prowazekii*, is transmitted in louse-infested environments and emerges in conditions of extreme poverty, war, and natural disaster. Patients experience a sudden onset of high fevers, severe headache, cough, myalgias, and abdominal pain. A maculopapular rash develops (primarily on the trunk) in more than half of patients and can progress to petechiae and purpura. Serious signs include delirium, coma, seizures, noncardiogenic pulmonary edema, skin necrosis, and peripheral gangrene. Mortality rates approached 60% in the preantibiotic era and continue to exceed 10–15% in contemporary outbreaks. *Scrub typhus*, caused by *Orientia tsutsugamushi* (a separate genus in the family Rickettsiaceae), is transmitted by larval mites or chiggers and is one of the most common infections in southeastern Asia and the western Pacific. The organism is found in areas of heavy scrub vegetation (e.g., along riverbanks). Patients may have an inoculation eschar and may develop a maculopapular rash, lymphadenopathy, and dyspnea. Severe cases progress to pneumonia, meningoencephalitis, myocarditis, DIC, and renal failure. Mortality rates range from 1% to 70% and vary by location, increasing age, myocarditis, delirium, pneumonitis, or signs of hemorrhage.

If recognized in a timely fashion, rickettsial disease is very responsive to treatment. Doxycycline (100 mg twice daily for 3–14 days) is the treatment of choice for both adults and children. Mortality rates are higher when tetracycline-based treatment is not given.

Purpura Fulminans (See also Chaps. 155 and 304) Purpura fulminans is the cutaneous manifestation of DIC and presents as large ecchymotic areas and hemorrhagic bullae. Progression of petechiae to purpura, ecchymoses, and gangrene is associated with congestive heart failure, septic shock, acute renal failure, acidosis, hypoxia, hypotension, and death. Purpura fulminans has been associated primarily with *N. meningitidis* but, in splenectomized patients, may be associated with *S. pneumoniae*, *H. influenzae*, and *S. aureus*.

Ecthyma Gangrenosum Septic shock caused by *P. aeruginosa* or, less often, *Aeromonas hydrophila* or other gram-negative organisms, can be associated with ecthyma gangrenosum **(see Figs. 164-1 and A1-34):** hemorrhagic vesicles surrounded by a rim of erythema with central necrosis and ulceration. Echthyma gangrenosum is most common among patients with neutropenia, extensive burns, and hypogammaglobulinemia.

Other Infections Associated with Rash *Vibrio vulnificus* and other noncholera *Vibrio* bacteremic infections **(Chap. 168)** can cause focal skin lesions and overwhelming sepsis in hosts with chronic liver disease, heavy alcohol consumption, iron storage disorders, diabetes, renal insufficiency, hematologic disease, or malignancy or other immunocompromising conditions. Over 95% of the cases are in the subtropical Pacific and Atlantic Oceans coastal regions in the Northern Hemisphere. After ingestion of contaminated raw shellfish (typically oysters from the Gulf Coast in U.S. cases), there is a sudden onset of malaise, chills, fever, and hypotension. The patient develops bullous or hemorrhagic skin lesions, usually on the lower extremities, and 75% of patients have leg pain. The mortality rate can be as high as 35–60%, particularly when the patient presents with hypotension. Outcomes are improved when patients are treated with fluoroquinolones with or without cephalosporins or with tetracycline-containing regimens. Other infections, caused by agents such as *Aeromonas*, *Klebsiella*, and *E. coli*, can cause hemorrhagic bullae and death due to overwhelming sepsis in cirrhotic patients. *Capnocytophaga canimorsus* can cause septic shock in asplenic or cirrhotic patients. Infection typically follows a dog bite. Serovars A–C appear more virulent, constituting 92% of human infections but only 7.6% of canine isolates. Patients present with fever, chills, myalgia, vomiting, diarrhea, dyspnea, confusion, and headache. Findings can include an exanthem or erythema multiforme **(see Figs. 56-9 and A1-24)**, cyanotic mottling or peripheral cyanosis, petechiae, and ecchymosis. About 30% of patients with this fulminant form die of overwhelming sepsis and DIC, and survivors may require amputation because of gangrene.

Erythroderma TSS **(Chaps. 147 and 148)** is usually associated with erythroderma. The patient presents with fever, malaise, myalgias, nausea, vomiting, diarrhea, and confusion. There is a sunburn-type rash that may be subtle and patchy but is usually diffuse and is found on the face, trunk, and extremities. Erythroderma, which desquamates after 1–2 weeks, is more common in *Staphylococcus*-associated than in *Streptococcus*-associated TSS. Hypotension develops rapidly—often within hours—after the onset of symptoms. Multiorgan failure occurs. Early renal failure may precede hypotension and distinguishes this syndrome from other septic shock syndromes. There may be no indication of a primary focal infection, although possible cutaneous or mucosal portals of entry for the organism can be ascertained when a careful history is taken. Colonization rather than overt infection of the vagina or a postoperative wound, for example, is typical with staphylococcal TSS, and the mucosal areas appear hyperemic but not infected. Streptococcal TSS is more often associated with skin or soft tissue infection (including necrotizing fasciitis), and patients are more likely to be bacteremic. TSS caused by *Clostridium sordellii* is associated with childbirth or with skin injection of black-tar heroin. The diagnosis of TSS is defined by the clinical criteria of fever, rash, hypotension, and multiorgan involvement. (Of note, fever is typically absent when TSS is caused by *C. sordellii*.) The mortality rate is 5% for

AFKEBOOKS
SINCE 2013

menstruation-associated TSS, 10–15% for nonmenstrual TSS, 30–70% for streptococcal TSS, and up to 90% for obstetric *C. sordellii* TSS. Clindamycin improves outcomes when included in the treatment regimen. Some studies have shown that use of IV immunoglobulin is associated with improved survival as well.

Viral Hemorrhagic Fevers Viral hemorrhagic fevers **(Chaps. 209 and 210)** are zoonotic illnesses caused by viruses that reside in either animal reservoirs or arthropod vectors. These diseases occur worldwide and are restricted to areas where the host species live. They are caused by four major groups of viruses: Arenaviridae (e.g., Lassa fever in Africa), Bunyaviridae (e.g., Rift Valley fever in Africa; hantavirus hemorrhagic fever with renal syndrome in Asia; and Crimean-Congo hemorrhagic fever, which has an extensive geographic distribution), Filoviridae (e.g., Ebola and Marburg virus infections in Africa), and Flaviviridae (e.g., yellow fever in Africa and South America and dengue in Asia, Africa, and the Americas). Lassa fever and Ebola and Marburg virus infections are also transmitted from person to person. The vectors for most viral fevers are found in rural areas; dengue and yellow fever are important exceptions. After a prodrome of fever, myalgias, and malaise, patients develop evidence of vascular damage, petechiae, and local hemorrhage. Shock, multifocal hemorrhaging, and neurologic signs (e.g., seizures or coma) predict a poor prognosis. Dengue **(Chap. 209)** is the most common arboviral disease worldwide. More than half a million cases of dengue hemorrhagic fever occur each year, with at least 12,000 deaths. Patients have a triad of symptoms: hemorrhagic manifestations, evidence of plasma leakage, and platelet counts of <100,000/μL. Mortality rates are 10–20%. If dengue shock syndrome develops, mortality rates can reach 40%. Ebola infection has been associated with outbreaks with high mortality rates. The 2014 outbreak in West Africa had a mortality rate of >50%. Symptoms can appear 2–21 days after exposure, but most patients become ill within 9 days. The patient first presents with fatigue, fever, headache, and muscle pains, and the illness can progress to multiorgan failure and hemorrhaging. Careful volume-replacement therapy to maintain blood pressure and intravascular volume is key to survival in these infections. Ribavirin also may be useful against Arenaviridae and Bunyaviridae.

Other viral illnesses with rash, such as measles, can be associated with significant mortality rates. Measles continues to be responsible for more than 100,000 deaths per year, worldwide, and to cause outbreaks in populations with low vaccination rates.

SEPSIS WITH A SOFT TISSUE/MUSCLE PRIMARY FOCUS

See also Chap. 129.

Necrotizing Fasciitis This infection is characterized by extensive necrosis of the subcutaneous tissue and fascia. It may arise at a site of minimal trauma or surgical incision and may also be associated with recent varicella, childbirth, or muscle strain. The most common causes of necrotizing fasciitis are group A streptococci alone **(Chap. 148)** and a mixed facultative and anaerobic flora **(Chap. 129)**; the incidence of group A streptococcal necrotizing fasciitis has been increasing for the past quarter-century. Diabetes mellitus, IV drug use, chronic liver or renal disease, and malignancy are associated risk factors. Physical findings are initially minimal compared with the severity of pain and the degree of fever. The examination is often unremarkable except for soft tissue edema and erythema. The infected area is red, hot, shiny, swollen, and exquisitely tender. In untreated infection, the overlying skin develops blue-gray patches after 36 h, and cutaneous bullae and necrosis develop after 3–5 days. Necrotizing fasciitis due to a mixed flora, but not that due to group A streptococci, can be associated with gas production. Without treatment, pain decreases because of thrombosis of the small blood vessels and destruction of the peripheral nerves—an ominous sign. The mortality rate is 15–34% overall, >70% in association with TSS, and nearly 100% without surgical intervention. With surgery, outcomes are significantly better; e.g., mortality is reduced to <40% in the setting of TSS. Necrotizing fasciitis may also be due to *Clostridium perfringens* **(Chap. 154)**; in this condition, the patient is extremely toxic and the mortality rate is high. Within 48 h, rapid tissue invasion and systemic toxicity associated with hemolysis and death ensue. The distinction between this entity and clostridial myonecrosis is made by muscle biopsy. Necrotizing fasciitis caused by community-acquired MRSA also has been reported.

Clostridial Myonecrosis (See also Chap. 154) Myonecrosis is often associated with trauma or surgery but can develop spontaneously. The incubation period is usually 12–24 h long, and massive necrotizing gangrene develops within hours of onset. Systemic toxicity, shock, and death can occur within 12 h. The patient's pain and toxic appearance are out of proportion to physical findings. On examination, the patient is febrile, apathetic, tachycardic, and tachypneic and may express a feeling of impending doom. Hypotension and renal failure develop later, and hyperalertness is evident preterminally. The skin over the affected area is bronze-brown, mottled, and edematous. Bullous lesions with serosanguineous drainage and a mousy or sweet odor can develop. Crepitus can occur secondary to gas production in muscle tissue. The mortality rate is >65% for spontaneous myonecrosis, which is often associated with *Clostridium septicum* or *C. tertium* and underlying malignancy. The mortality rates associated with trunk and limb infection are 63% and 12%, respectively, and any delay in surgical treatment increases the risk of death.

NEUROLOGIC INFECTIONS WITH OR WITHOUT SEPTIC SHOCK

Bacterial Meningitis (See also Chap. 138) Bacterial meningitis is one of the most common infectious disease emergencies involving the central nervous system. Although hosts with cell-mediated immune deficiency (including transplant recipients, diabetic patients, elderly patients, and cancer patients receiving certain chemotherapeutic agents) are at particular risk for *Listeria monocytogenes* meningitis, most cases in adults are due to *S. pneumoniae* (30–60%) and *N. meningitidis* (10–35%). The classic presentation of fever, meningismus, and altered mental status is seen in only one-half to two-thirds of patients and less than one-half of adults. The elderly can present without fever or meningeal signs. Cerebral dysfunction is evidenced by confusion, delirium, and lethargy that can progress to coma. In some cases, the presentation is fulminant, with sepsis and brain edema; papilledema at presentation is unusual and suggests another diagnosis (e.g., an intracranial lesion). Focal signs, including cranial nerve palsies (IV, VI, VII), can be seen in 10–20% of cases; 50–70% of patients have bacteremia. A poor outcome is associated with coma, seizures, hypotension, a pneumococcal etiology, respiratory distress, a CSF glucose level of <0.6 mmol/L (<10 mg/dL), a CSF protein level of >2.5 g/L, a peripheral white blood cell count of <5000/μL, and a serum sodium level of <135 mmol/L. Rapid initiation of treatment is essential; the odds of an unfavorable outcome may increase by 30% for each hour that treatment is delayed. Dexamethasone is an adjunctive treatment for meningitis in adults, especially for infections caused by *S. pneumoniae*. It must be given before or with the first dose of antibiotics; otherwise, it is unlikely to improve outcomes.

Suppurative Intracranial Infections (See also Chap. 140) Suppurative intracranial infections present along with sepsis and hemodynamic instability. Rapid recognition of the toxic patient with central neurologic signs is crucial to improvement of the prognosis of these entities. Patients with diabetes or hematologic disease may be at increased risk for these infections. *Subdural empyema* arises from the paranasal sinus in 60–70% of cases. Microaerophilic streptococci and staphylococci are the predominant etiologic organisms. The patient is toxic, with fever, headache, and nuchal rigidity. Of all patients, 75% have focal signs and 6–20% die. Despite improved survival rates, 15–44% of patients are left with permanent neurologic deficits. *Septic cavernous sinus thrombosis* follows a facial or sphenoid sinus infection; 70% of cases are due to staphylococci (including MRSA), and the remainder are due primarily to aerobic or anaerobic streptococci. Fungi have been common in some series. A unilateral or retro-orbital headache progresses to a toxic appearance and fever within days. Three-quarters of patients have unilateral periorbital edema that

AFKEBOOKS
SINCE 2013

becomes bilateral and then progresses to ptosis, proptosis, ophthalmoplegia, and papilledema. The mortality rate is as high as 30% in older studies. Recent reports indicate improved survival as high as 90% with fewer sequelae. *Septic thrombosis of the superior sagittal sinus* spreads from the ethmoid or maxillary sinuses and is caused by *S. pneumoniae*, other streptococci, and staphylococci. The fulminant course is characterized by headache, nausea, vomiting, rapid progression to confusion and coma, nuchal rigidity, and brainstem signs. If the sinus is totally thrombosed, the mortality rate exceeds 80%. Broad-spectrum antibiotics and early surgical intervention at the primary site of infection may improve outcomes. Anticoagulation or steroids are of uncertain benefit.

Brain Abscess **(See also Chap. 140)** Brain abscess often occurs without systemic signs. Almost half of patients are afebrile, and presentations are more consistent with a space-occupying lesion in the brain; 70% of patients have headache and/or altered mental status, 50% have focal neurologic signs, and 25% have papilledema. Abscesses can present as single or multiple lesions resulting from contiguous foci or hematogenous infection, such as endocarditis, or after surgery or trauma. The infection progresses over several days from cerebritis to an abscess with a mature capsule. More than half of infections are polymicrobial, with an etiology consisting of aerobic bacteria (primarily streptococcal species) and anaerobes. Abscesses arising hematogenously are especially apt to rupture into the ventricular space, causing a sudden and severe deterioration in clinical status and a high mortality rate. Otherwise, mortality is low (<20%) but morbidity is high (30–55%). Patients presenting with stroke and a parameningeal infectious focus, such as sinusitis or otitis, may have a brain abscess, and physicians must maintain a high level of suspicion. Prognosis worsens in patients with a fulminant course, delayed diagnosis, abscess rupture into the ventricles, multiple abscesses, or abnormal neurologic status at presentation. In one study, mortality at 1 year was 19%.

Cerebral Malaria **(See also Chap. 224)** This entity should be urgently considered for inhabitants of, or recent travelers to, areas endemic for malaria. Patients present with a febrile illness and lethargy or other neurologic signs. Fulminant malaria is caused by *Plasmodium falciparum* and is associated with temperatures of >40°C (>104°F), hypotension, jaundice, acute respiratory distress syndrome, and bleeding. By definition, any patient with a change in mental status or repeated seizure in the setting of fulminant malaria has cerebral malaria. In adults, this nonspecific febrile illness progresses to coma over several days; occasionally, coma occurs within hours and death within 24 h. Nuchal rigidity and photophobia are rare. On physical examination, symmetric encephalopathy is typical, and upper motor neuron dysfunction with decorticate and decerebrate posturing can be seen in advanced disease. Unrecognized infection results in a 20–30% mortality rate. Children with neurologic deficit at hospital discharge, seizure recurrence during treatment, and/or ischemic neural injury on MRI are at particular risk for neurologic and mental health sequelae.

Intracranial and Spinal Epidural Abscesses **(See also Chap. 442)** Spinal and intracranial epidural abscesses (SEAs and ICEAs) can result in permanent neurologic deficits, sepsis, and death. At-risk patients include those with diabetes mellitus; IV drug use; chronic alcohol abuse; recent spinal trauma, surgery, or epidural anesthesia; and other comorbid conditions, such as HIV infection. Fungal epidural abscess and meningitis can follow epidural or paraspinal glucocorticoid injections. In the United States and Canada, where early treatment of otitis and sinusitis is typical, ICEA is rare but the number of cases of SEA is on the rise. In areas with limited access to health care, SEAs and ICEAs cause significant morbidity and mortality. ICEAs typically present as fever, mental status changes, and neck pain, while SEAs often present as fever, localized spinal tenderness, and back pain. ICEAs are typically polymicrobial, whereas SEAs are most often due to hematogenous seeding, with staphylococci the most common etiologic agent. Early diagnosis and treatment, which may include surgical drainage, minimize rates of mortality and permanent neurologic sequelae. Outcomes are worse for SEA due to MRSA, for infection at a higher vertebral-body level, for impaired neurologic status on presentation, and for dorsal rather than ventral abscess location. Elderly patients and persons with renal failure, malignancy, and other comorbidities also have less favorable outcomes.

OTHER FOCAL SYNDROMES WITH A FULMINANT COURSE

Infection at virtually any primary focus (e.g., osteomyelitis, pneumonia, pyelonephritis, or cholangitis) can result in bacteremia and sepsis. Lemierre's syndrome—jugular septic thrombophlebitis caused by *Fusobacterium necrophorum*—is associated with metastatic infectious emboli (primarily to the lung but sometimes to the liver or other organs) and sepsis, with mortality rates of >15%. Fusobacterium bloodstream infections have been associated with occult gastrointestinal or genitourinary malignancy. TSS has been associated with focal infections such as septic arthritis, peritonitis, sinusitis, and wound infection. Rapid clinical deterioration and death can be associated with destruction of the primary site of infection, as is seen in endocarditis and in infections of the oropharynx (e.g., Ludwig's angina or epiglottitis, in which edema suddenly compromises the airway).

Rhinocerebral Mucormycosis **(See also Chap. 218)** Individuals with diabetes or immunocompromising conditions such as solid organ transplants or hematologic malignancies are at risk for invasive rhinocerebral mucormycosis. Patients present with low-grade fever, dull sinus pain, diplopia, decreased mental status, decreased ocular motion, chemosis, proptosis, dusky or necrotic nasal turbinates, and necrotic hard-palate lesions that respect the midline. Without rapid recognition, surgical intervention and antifungal therapy, the process continues on an inexorable invasive course, with mortality rates of 50–85% or greater. Uncontrolled diabetes and increasing age are negative prognostic factors.

Acute Bacterial Endocarditis **(See also Chap. 128)** This entity presents with a much more aggressive course than subacute endocarditis. Bacteria such as *S. aureus*, *S. pneumoniae*, *L. monocytogenes*, *Haemophilus* species, and streptococci of groups A, B, and G attack native valves. Native-valve endocarditis caused by *S. aureus* (including MRSA strains) is increasing. Mortality rates range from 10% to 40%. The host may have comorbid conditions such as underlying malignancy, diabetes mellitus, IV drug use, or alcoholism. The patient presents with fever, fatigue, and malaise <2 weeks after onset of infection. On physical examination, a changing murmur and congestive heart failure may be noted. Hemorrhagic macules on palms or soles (*Janeway lesions*) sometimes develop. Petechiae, Roth's spots, splinter hemorrhages, and splenomegaly are unusual. Rapid valvular destruction, particularly of the aortic valve, results in pulmonary edema and hypotension. Myocardial abscesses can form, eroding through the septum or into the conduction system and causing life-threatening arrhythmias or high-degree conduction block. Large friable vegetations can result in major arterial emboli, metastatic infection, or tissue infarction. Older patients with *S. aureus* endocarditis are especially likely to present with nonspecific symptoms—a circumstance that delays diagnosis and worsens prognosis. Rapid intervention is crucial for a successful outcome.

Inhalational Anthrax **(See also Chap. S3)** Inhalational anthrax, the most severe form of disease caused by *B. anthracis*, had not been reported in the United States for more than 25 years until the use of this organism as an agent of bioterrorism in 2001. Patients presented with malaise, fever, cough, nausea, drenching sweats, shortness of breath, and headache. Rhinorrhea was unusual. All patients had abnormal chest roentgenograms at presentation. Pulmonary infiltrates, mediastinal widening, and pleural effusions were the most common findings. Hemorrhagic meningitis was documented in 38% of these patients. Survival was more likely when antibiotics were given during the prodromal period and when multidrug regimens were used. In the absence of urgent intervention with antimicrobial agents and supportive care, inhalational anthrax progresses rapidly to hypotension, cyanosis, and death.

AFKEBOOKS
SINCE 2013

Viral Respiratory Tract Illness Viral respiratory tract illnesses can cause severe disease; several new syndromes have been described in the past decade. For patients who present with a respiratory illness and a relevant exposure and travel history, these viral illnesses must be considered and appropriate infection control measures instituted in addition to supportive care.

Avian and Swine Influenza **(See also Chap. 200)** Human cases of avian influenza have occurred primarily in Southeast Asia, particularly Vietnam (H5N1) and China (H7N9). Avian influenza should be considered in patients with severe respiratory tract illness, particularly if they have been exposed to poultry. Patients present with high fever, an influenza-like illness, and lower respiratory tract symptoms; this illness can progress rapidly to bilateral pneumonia, acute respiratory distress syndrome, multiorgan failure, and death. Younger age appears to be associated with a lower risk of complications. Early antiviral treatment with neuraminidase inhibitors should be initiated along with aggressive supportive measures. Unlike avian influenza, whose human-to-human transmission has so far been rare and has not been sustained, influenza caused by a novel swine-associated A/H1N1 virus has spread rapidly throughout the world; by 2012, 214 countries had diagnosed cases of influenza A/H1N1, with 18,449 deaths. Patients most at risk of severe disease are children <5 years of age, elderly persons, patients with underlying chronic conditions, and pregnant women. Obesity also has been identified as a risk factor for severe illness. Immunosuppression and co-infection with *S. aureus* at presentation are independent risk factors for increased mortality.

SARS, COVID-19, and MERS **(See Chap. 199)** Severe acute respiratory syndrome (SARS) caused by what is now labeled SARS-CoV-1 was identified in 2002 in China but has been diagnosed in several countries, primarily in Asia. Possible animal reservoirs include bats and civets. SARS-CoV-1 is characterized by efficient human transmission but relatively low mortality. It spreads from person to person via droplets; "super-spreader" airborne events have occurred. The potential SARS-CoV-1 pandemic was controlled through identification and isolation of infected patients. A 3- to 7-day prodrome characterized by fever, malaise, headache, and myalgia can progress to nonproductive cough, dyspnea, and respiratory failure. The risk of contagion is low during the prodrome. Older patients and those with diabetes mellitus, chronic hepatitis B, and other comorbidities can have less favorable outcomes.

COVID-19 caused by SARS-CoV-2 has resulted in a pandemic of historic proportion. Originally linked to cases in China, SARS-CoV-2 has spread internationally at a rapid pace. As of June 2021, cases approached 174 million worldwide with 3.7 million deaths; the United States had more than 33 million cases and ~592,000 deaths. Actual infection may exceed reported cases by 8–10 fold. The primary mode of transmission of COVID-19 is through direct person-to-person contact via respiratory droplets; transmission by the airborne route or by contact with contaminated surfaces has been documented but is much less common. Greatest risk of infection is associated with close and prolonged contact such as in household or congregate settings. Rates of secondary infection among household contacts are high, over 50% in recent studies.

Although ~80% of patients with symptomatic COVID-19 infection have mild disease such as cough, fever, gastrointestinal symptoms, and taste and smell alterations, ~5% of patients develop respiratory failure, shock, and multiorgan system failure. Patients typically have been symptomatic for 5–7 days before progression to severe pneumonia and hypoxemia. Shock, often in the setting of cardiac injury/myocarditis, arrhythmias or cardiomyopathy, and thromboembolic complications such as pulmonary emboli or stroke, can occur. Other manifestations include acute kidney or liver injury. Laboratory findings include lymphopenia, elevated lactate dehydrogenase, and often evidence of a cytokine-release type syndrome with elevated C-reactive protein, D-dimer, ferritin, and proinflammatory cytokines such as interleukin 6. Chest radiographs and CT imaging reveal diffuse ground-glass opacities. Consolidation, predominantly in the lower lobes and peripherally, is a frequent finding. Mortality rates for patients requiring intensive care and mechanical ventilation have improved significantly over the course of the pandemic but remain ≥20%. Risk factors for severe disease and poor outcomes include age ≥65 years, morbid obesity, diabetes, cardiovascular or cerebrovascular disease, hypertension, chronic obstructive lung disease, and chronic kidney disease. Severe disease and mortality is also increased in males and pregnant women.

Treatment currently includes supportive care, antiviral therapy, anti-inflammatory therapy, and anticoagulation prophylaxis or treatment. Patients with respiratory distress are hypoxemic but maintain pulmonary mechanical functioning; thus, invasive mechanical ventilation should be delayed until other interventions have been exhausted. In a meta-analysis, dexamethasone or other glucocorticoids reduced 28-day mortality in patients with severe COVID-19 disease compared with standard care: 32% and 40%, respectively (odds ratio, 0.66; 95% confidence interval, 0.53–0.82). Secondary bacterial pneumonia is uncommon in COVID-19, and antibacterial agents should not be routinely prescribed. With the availability of highly effective vaccines, prevention is the most important strategy to control COVID-19.

Middle East respiratory syndrome (MERS) is caused by a novel betacoronavirus with a likely bat reservoir and was first recognized in 2012 in Saudi Arabia. Human cases have been associated with direct and indirect contact with dromedary camels. Unlike SARS and COVID-19, MERS exhibits inefficient human transmission but carries a high mortality rate. As of 2017, >2000 cases had been confirmed, most in the Arabian Peninsula, with 35% mortality. MERS ranges from asymptomatic infection to acute respiratory distress syndrome, multiorgan failure, and death. Elderly men with comorbidities appear to be at highest risk for poor outcomes. Despite little documented human-to-human transmission in the community, nosocomial infection must be prevented by adherence to strict infection control practices. MERS is currently a low-level public health threat and is likely to remain so unless the virus mutates and its transmissibility increases.

Hantavirus Pulmonary Syndrome **(See also Chap. 209)** Hantavirus pulmonary syndrome has been documented in the United States since 1993 (primarily the southwestern states, west of the Mississippi River), Canada, and South America. Most cases occur in rural areas and are associated with exposure to rodents. Patients present with a nonspecific viral prodrome of fever, malaise, myalgias, nausea, vomiting, and dizziness that may progress to pulmonary edema, respiratory failure, and death. Hantavirus pulmonary syndrome causes myocardial depression and increased pulmonary vascular permeability; therefore, careful fluid resuscitation and use of pressor agents are crucial. Aggressive cardiopulmonary support during the first few hours of illness can be life-saving in this high-mortality syndrome. The early onset of thrombocytopenia may help distinguish this syndrome from other febrile illnesses in an appropriate epidemiologic setting.

***Clostridioides difficile* Infection** *C. difficile* infection (CDI) is a toxin-mediated diarrheal syndrome that is strongly associated with prior antibiotic use. Proton-pump inhibitors also have been identified as a potential risk factor for the disease. Although most cases of CDI have occurred in the health care setting, community-onset CDI is increasing. Overall, community-onset cases occur in younger patients than nosocomial cases. Patients with community-onset CDI are less likely to have a history of antibiotic or protein-pump inhibitor use. CDI is associated with significant morbidity and mortality, particularly among older patients. The CDC named *C. difficile* infection as one of the top three health threats associated with antibiotic use.

SUMMARY

Acutely ill febrile patients with the syndromes discussed in this chapter require close observation, aggressive supportive measures, and—in most cases—admission to intensive care units. The most important task of the physician is to distinguish these patients from other infected febrile patients whose illness will not progress to fulminant disease. The alert physician must recognize the acute infectious disease emergency and proceed with appropriate urgency.

AFKEBOOKS
SINCE 2013

FURTHER READING

KRAUSE PJ: Human babesiosis. Int J Parasitol 49:165, 2019.

NELSON GE et al: Epidemiology of invasive Group A *Streptococcal* infections in the United States, 2005–2012. Clin Infect Dis 63:478, 2016.

RHEE C et al: Incidence and trends of sepsis in US hospitals using clinical vs claims data, 2009–2014. JAMA 318:1241, 2017.

STERNE JA et al: Association between administration of systemic corticosteroids and mortality among critically ill patients with COVID-19: A meta-analysis. JAMA 324:1330, 2020.

THEILACKER C et al: Overwhelming postsplenectomy infection: A prospective multicenter cohort study. Clin Infect Dis 62:871, 2016.

ZUNT JR et al: Global, regional, and national burden of meningitis, 1990–2016: A systematic analysis for the Global Burden of Disease Study 2016. Lancet Neurol 17:1061, 2018.

123 Immunization Principles and Vaccine Use

Sarah Mbaeyi, Amanda Cohn, Nancy Messonnier

Few medical interventions of the past century can rival the effect that immunization has had on longevity, economic savings, and quality of life. Seventeen diseases are now preventable through vaccines routinely administered to children and adults in the United States (Table 123-1), and most vaccine-preventable diseases of childhood are at historically low levels (Table 123-2). Health care providers deliver the vast majority of vaccines in the United States in the course of providing routine health services and therefore play an integral role in the nation's public health system.

The COVID-19 pandemic has highlighted the importance of a strong immunization program: robust surveillance systems to detect emerging infectious disease threats; public-private partnerships to accelerate the development of novel vaccines; and systems in place to rapidly implement a vaccination program and monitor vaccine safety and effectiveness. As of July 2021, multiple COVID-19 vaccines have received authorization for emergency use in the United States and over two-thirds of adults have received at least one dose. Furthermore, the largest and most comprehensive vaccine safety program in U.S. history has demonstrated the safety of COVID-19 vaccines, and evaluations have demonstrated the vaccines to be effective against SARS-CoV-2 infections, including against severe disease that would otherwise result in hospitalization and death.

TABLE 123-1 Diseases Preventable with Vaccines Routinely Administered in the United States to Children and/or Adults

CONDITION	TARGET POPULATION(S) FOR ROUTINE USE
Pertussis	Children, adolescents, adults
Diphtheria	Children, adolescents, adults
Tetanus	Children, adolescents, adults
Poliomyelitis	Children
Measles	Children
Mumps	Children
Rubella, congenital rubella syndrome	Children
Hepatitis B	Children and high-risk adults
Haemophilus influenzae type b infection	Children and high-risk adults
Hepatitis A	Children and high-risk adults
Influenza	Children, adolescents, adults
Varicella	Children
Pneumococcal disease	Children, older adults, and high-risk adults[a]
Serogroups A, C, W, Y meningococcal disease	Adolescents and high-risk children and adults
Serogroup B meningococcal disease	High-risk children and adults[a]
Rotavirus infection	Infants
Human papillomavirus infection, cervical and anogenital cancers	Adolescents and young adults[a]
Zoster	Older adults

[a]Others in certain age groups may be vaccinated based on shared clinical decision-making.

TABLE 123-2 Decline in Vaccine-Preventable Diseases in the United States Following Widespread Implementation of National Vaccine Recommendations

CONDITION	ANNUAL NO. OF PREVACCINE CASES (AVERAGE)	NO. OF CASES REPORTED IN 2019[a]	REDUCTION (%) IN CASES AFTER WIDESPREAD VACCINATION
Smallpox	29,005	0	100
Diphtheria	21,053	2	>99
Measles	530,217	1287	>99
Mumps	162,344	3509	98
Pertussis	200,752	15,662	92
Polio (paralytic)	16,316	0	100
Rubella	47,745	3	>99
Congenital rubella syndrome	152	0	100
Tetanus	580	19	97
Haemophilus influenzae type b infection	20,000	14[b]	>99
Hepatitis A	117,333	4000[c]	97
Hepatitis B (acute)	66,232	20,900[c]	68
Invasive pneumococcal infection: all ages	63,067	30,400[d]	52
Rotavirus hospitalizations (<3 years old)	62,500	30,625[e]	51
Varicella	4,085,120	102,128[f]	98

[a]2019 reported cases unless otherwise specified (provisional as of January 2020). [b]An additional 12 type b infections are estimated to have occurred among 243 reports of *H. influenzae* infection caused by unknown types among children <5 years of age. [c]Data from the CDC's Viral Hepatitis Surveillance, 2016. [d]Unpublished data from the CDC's Active Bacterial Core Surveillance, 2016. [e]Unpublished data from the CDC's New Vaccine Surveillance Network, 2017, U.S. rotavirus disease now has biennial pattern. [f]Unpublished data from CDC's varicella program, 2017.

Source: Adapted from SW Roush et al: JAMA 298:2155, 2007 and Morb Mortal Wkly Rep 65:924, 2017.

VACCINE IMPACT

Direct and Indirect Effects Immunizations against specific infectious diseases protect individuals against infection and thereby prevent symptomatic illnesses. In addition, specific vaccines may also blunt the severity of clinical illness (e.g., rotavirus vaccines and severe gastroenteritis) or reduce complications (e.g., zoster vaccines and postherpetic neuralgia). Some immunizations also reduce transmission of infectious disease agents from immunized people to others, thereby reducing the impact of infection spread. This indirect impact is known as *herd immunity*. The level of immunization in a population that is required to achieve indirect protection of unimmunized people varies substantially with the specific vaccine and disease.